

Gatekeeper Model v2: Quantitative Parameters by Layer

Layer 1: Landscape Remodeling

Parameter	Value
4mC promoter-expression correlation	$\rho = 0.139$, $p = 8.26 \times 10^{-4}$
Permutation test (4mC T2vsT1)	$r = 0.125$, $p = 0.002$
CCGG 4mC positional overlap	Jaccard = 0.000
AAGCCCG site loss (SC_RS17645↓)	260 sites, 0% retention
T1 CCGG core genome fraction	77%
T2 CCGG chromosomal arms	89%
6mA T2vsT1 correlation	Lost after T3 resequencing

Layer 2: Protection / Depletion

Parameter	Value
TF binding site methylation	fold = 0.66, $p = 6.8 \times 10^{-5}$
σ -10 box methylation depletion	$p = 1.73 \times 10^{-12}$
Literature 37 TFs methylated	3/37 (8%), all constitutive
SARP activators methylated	0/4 (0%)
SARP activator LFC range	+2.0 to +7.2 (unmethylated)
DMG functional enrichment	NONE (GO/KEGG/COG)
TF BS methylation: Lost T2T3	75%

Layer 3: Signal Gating (Revised)

Parameter	Value
Total regulatory genes screened	1,055 (25 families)
Methylated regulatory genes	165/1,055 (15.6%)
Coordinated methyl+expr genes	62 (all outside literature 37)
T3 discordant_gain_up at arms	87% (OR=8.05, $p=0.001$)
COG T (signal transduction)	OR=2.47, $p=0.045$
COG Q (secondary metabolism)	0 genes
TCS asymmetric pairs	7 identified
Top candidate: SC_RS10435	Chaplin-adjacent TetR
Top candidate: SC_RS35525	PPTase-adjacent SK

Regulatory Family Methylation Rates

